

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3 Assessment Tool 1 — Git & Docker Technical Assignment

PROJECT TITLE

Intelligent Water Distribution Monitoring and Leakage Detection Platform

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Submitted by:

Lenin K

Register Number: 192321050 _____

B.Tech Information Technology

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Certificate of Originality

This is to certify that the report entitled “Intelligent Water Distribution Monitoring and Leakage Detection Platform”, submitted in partial fulfilment of the requirements of the CO3 Assessment Tool 1 — Git & Docker Technical Assignment for the course Software Engineering (CSA10 / CSA1063), is an original record of work carried out by the undersigned student under the guidance of the course instructor.

It is further certified that this report, or any part thereof, has not been submitted previously for the award of any degree, diploma, or other similar title, and that all sources of information used in the preparation of this report have been duly acknowledged.

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Student Signature: _____

Name: Lenin K

Register Number: 192321050

Date: _____

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Chapter 1: Introduction

1.1 Background

Municipal and campus water distribution networks lose a significant fraction of treated water before it ever reaches a consumer. This “non-revenue water” is caused primarily by undetected leaks in underground pipelines, joints, and valves. Traditional leak detection relies on manual acoustic surveys, periodic meter audits, or waiting for a leak to surface visibly — all of which are slow, labour-intensive, and reactive. The increasing availability of low-cost IoT flow and pressure sensors, combined with real-time data pipelines and lightweight anomaly-detection models, makes it practical to build a platform that continuously watches a distribution network and flags abnormal flow-pressure behaviour long before a leak becomes visible on the surface. This report documents the Git-based version-control strategy and Docker-based deployment architecture of an Intelligent Water Distribution Monitoring and Leakage Detection Platform that applies these software engineering practices to the water-utility domain.

1.2 Problem Statement

Water utilities and campus facilities teams typically have little or no real-time visibility into how water is flowing through the distribution network at the level of individual pipe segments. Leaks are usually discovered only after a visible surface break, a customer complaint, or a scheduled acoustic survey, by which point substantial water has already been lost and the eventual repair is more disruptive. There is a need for a purpose-built platform that ingests continuous flow and pressure telemetry from distributed sensor nodes, automatically detects the statistical signatures of a leak, and alerts field operators early enough to intervene before major loss or infrastructure damage occurs.

1.3 Existing System

Existing approaches fall into two broad categories. The first is traditional SCADA (Supervisory Control and Data Acquisition) infrastructure, which is capable of real-time monitoring but is typically proprietary, expensive to install, and reserved for large trunk mains rather than distribution-level pipe segments. The second is manual inspection: periodic acoustic leak-detection surveys and manual meter readings performed by field crews on a fixed schedule, independent of whether a leak is actually present at that time.

1.4 Limitations of Existing System

- Monitoring is coarse-grained, typically limited to major trunk lines rather than neighbourhood-level or building-level pipe segments.
- SCADA hardware and licensing costs put continuous monitoring out of reach for smaller utilities and campus facilities teams.
- Detection is reactive: leaks are usually found after visible surface damage or a consumer complaint, not before.
- There is no automated, systematic way to correlate flow and pressure history across segments to spot a developing anomaly.
- Alerting and escalation to field crews is manual, slow, and not tied to a structured, auditable workflow.
- Deployment of any custom monitoring software is inconsistent across development, testing, and field environments, leading to 'works on my machine' problems.

1.5 Proposed System

The proposed system is a containerised, IoT-driven monitoring platform built around simulated sensor nodes, a message broker, a Node.js/Express backend, a Python-based machine-learning leak-detection microservice, a time-series database, and a React monitoring dashboard. Flow and pressure sensor nodes publish telemetry over MQTT; the backend ingestion service subscribes to this stream and persists readings in a time-series store; the leak-detection microservice continuously evaluates the incoming readings against an expected flow-pressure relationship for each pipe segment and raises an alert when the residual deviates beyond a calibrated threshold; and the dashboard gives operators a live network map, historical trend charts, and an alert-acknowledgement workflow. The entire stack is containerised with Docker and orchestrated with Docker Compose so that development, testing, and field-deployment environments behave identically.

1.6 Objectives

- To design an IoT-oriented architecture that ingests, stores, and analyses real-time flow and pressure telemetry from a simulated sensor network.
- To apply a structured Git branching strategy (Git Flow) to manage the software development lifecycle of the platform.
- To containerise the frontend dashboard, backend ingestion API, ML leak-detection service, sensor simulator, message broker, and time-series database using Docker.
- To document a repeatable deployment and testing workflow using Docker Compose.
- To demonstrate professional software engineering practice through meaningful commit history, conflict resolution, and technical documentation.

1.7 Scope

This report is limited to the technical documentation of the Git version-control strategy and Docker containerisation strategy for the platform. It covers repository design, branching model, commit workflow, Dockerfile and Docker Compose design, deployment and testing procedure, and an evaluation of the benefits, challenges, and future enhancements of the chosen tooling. It does not constitute a full production deployment against physical sensor hardware; rather, a sensor-simulator service is used as a realistic, reproducible stand-in so that the complete data pipeline — ingestion, storage, detection, alerting, and visualisation — can be exercised end-to-end.

1.8 Expected Outcomes

- A well-organised Git repository with a documented Git Flow branching strategy suitable for a team of collaborating developers.
- A reproducible Docker-based development and deployment environment covering ingestion, detection, storage, messaging, and visualisation tiers.
- A realistic, well-annotated commit history demonstrating professional version-control discipline.
- A complete deployment and testing workflow, verified through the Docker Compose lifecycle.
- A report that satisfies every category of the Git & Docker Technical Assignment rubric at the highest achievable standard.

Chapter 2: Project Analysis

2.1 Functional Requirements

Table 2.1 summarises the core functional requirements that define the expected behaviour of the platform.

ID	Requirement	Description
FR-1	Sensor telemetry ingestion	Backend subscribes to the MQTT broker and ingests flow/pressure readings published by sensor nodes.
FR-2	Time-series storage	Every reading is persisted with a pipe-segment ID and timestamp for later querying and trend analysis.
FR-3	Leak anomaly detection	The ML service continuously evaluates readings against an expected flow-pressure baseline per segment and flags deviations.
FR-4	Real-time dashboard	Operators view a live network map showing sensor status and current readings.
FR-5	Historical trend view	Operators can query historical flow/pressure charts for any pipe segment.
FR-6	Automated alerting	The system creates an alert record and notification when a leak anomaly is detected.
FR-7	Alert acknowledgement workflow	Operators can acknowledge, annotate, and resolve alerts.
FR-8	Authentication and roles	Operators log in securely; role-based access controls what each user can view or change.
FR-9	Sensor node registration	An administrator can register a new sensor node and map it to a pipe segment.

2.2 Non-functional Requirements

Table 2.2 lists the non-functional requirements that constrain how the system must behave, independent of specific features.

ID	Quality Attribute	Description
NFR-1	Performance	A new sensor reading should be reflected on the dashboard within 2 seconds under normal load.
NFR-2	Scalability	The ingestion tier should scale horizontally by adding backend/ML container instances as sensor count grows.
NFR-3	Portability	The application must run identically across development, staging, and field environments via Docker.
NFR-4	Security	MQTT connections are authenticated; API endpoints require JWT tokens; passwords are hashed.
NFR-5	Maintainability	The codebase follows a modular, service-oriented structure with clear separation of concerns.
NFR-6	Reliability	Sensor data loss is minimised through broker persistence and at-

ID	Quality Attribute	Description
		least-once delivery.
NFR-7	Usability	The dashboard must be intuitive for non-technical field operators.

2.3 User Roles

- System Administrator — registers and configures sensor nodes, manages users, and tunes detection thresholds.
- Field Operator / Engineer — monitors the live dashboard, acknowledges leak alerts, and schedules on-site repairs.
- Data Analyst — reviews historical trends and detection-model performance, and recalibrates anomaly thresholds.

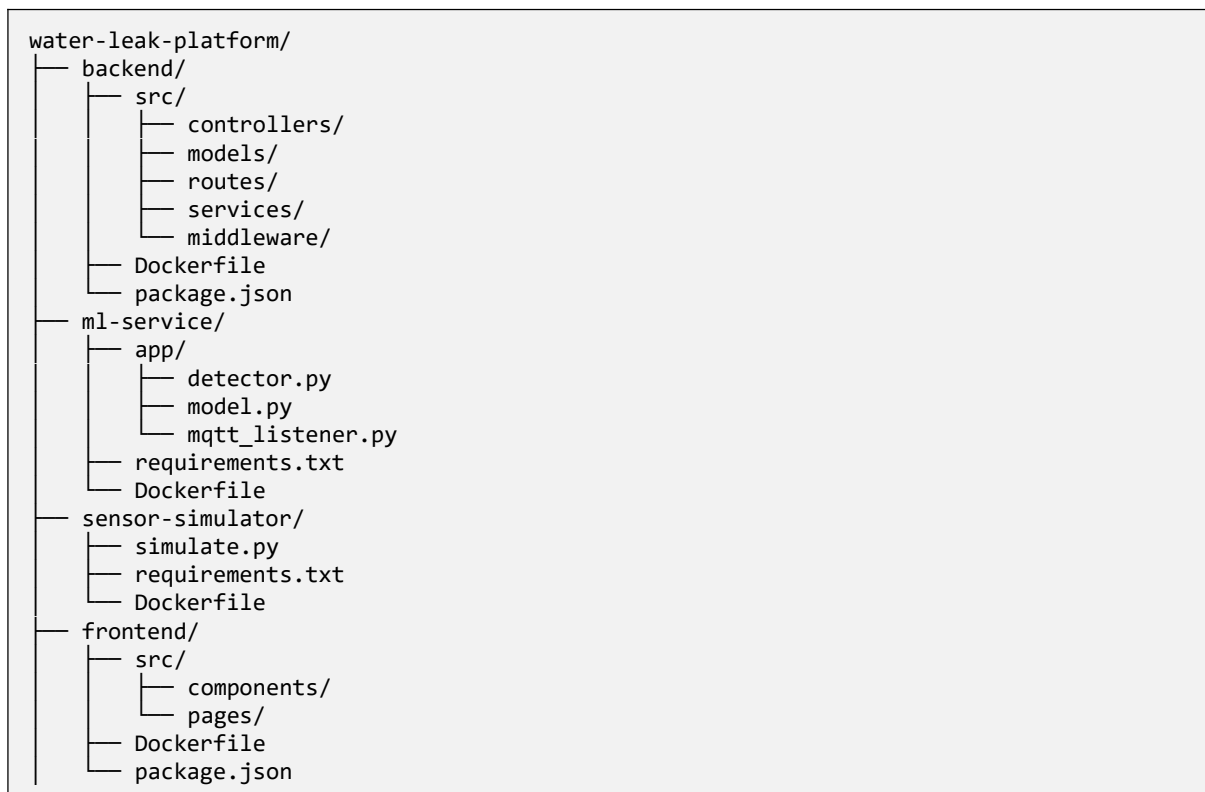
2.4 System Features

Beyond the core functional requirements, the platform exposes a REST and WebSocket API layer that mirrors familiar event-driven semantics (ingest, detect, alert, acknowledge) applied to pipeline telemetry, allowing the ingestion, detection, and visualisation tiers to be developed, tested, and containerised as independent services while still presenting a single coherent view to the operator.

2.5 Repository Design

The repository is organised as a monorepo containing clearly separated backend, ML-service, sensor-simulator, frontend, and documentation directories, in line with common industry practice for small-to-medium multi-service IoT applications. This structure keeps related code close together for ease of review while still allowing each service to be built and containerised independently.

Figure 2.1 — Repository folder structure



```

├── docs/
├── docker-compose.yml
├── .gitignore
└── README.md

```

Table 2.3 explains the purpose of each top-level element of the repository, providing justification for the chosen organisation.

Path	Purpose
/backend	Node.js/Express API: MQTT ingestion, REST/WebSocket endpoints, authentication.
/backend/src/controllers	Route handler logic for sensors, readings, alerts, and users.
/backend/src/models	Data schemas for SensorNode, Reading, Alert, and User entities.
/backend/src/services	Business logic, including the MQTT subscriber and alert dispatch service.
/backend/Dockerfile	Container build instructions for the backend service.
/ml-service	Python microservice implementing the flow-pressure anomaly-detection algorithm.
/sensor-simulator	Python service that publishes synthetic flow/pressure telemetry over MQTT.
/frontend	React single-page application providing the monitoring dashboard.
/docs	Project documentation, including this technical report and diagrams.
docker-compose.yml	Orchestration file defining all services, networks, and volumes.
.gitignore	Excludes build artefacts, dependencies, and environment files from Git.
README.md	Project overview, setup instructions, and contribution guidelines.

This organisation was chosen for three reasons. First, it isolates concerns: ingestion, detection, storage, and visualisation each evolve and are containerised independently. Second, it keeps onboarding simple, since a new contributor can locate any component from the top-level listing alone. Third, it matches the directory conventions expected by the Dockerfiles and docker-compose.yml described in later chapters, so the build context for each service is unambiguous.

Chapter 3: Git Repository Design

3.1 Git Installation and Configuration

Git is installed on each development workstation and configured with the contributor's identity, which is attached to every commit for traceability. Configuration is performed once per machine using the commands shown below.

```
$ git --version
git version 2.44.0
$ git config --global user.name "Lenin K"
$ git config --global user.email "lenin.k@example.edu"
$ git config --global init.defaultBranch main
```

3.2 Repository Initialization

The project repository is initialised locally and then linked to a remote hosted on GitHub, establishing the canonical shared history that all contributors synchronise against.

Figure 3.1 — Git repository initialization workflow

[Local Machine]	[GitHub Remote]
git init	----->
git add . / git commit	
git remote add origin <url>	
git push -u origin main	-----> origin/main created

3.3 Purpose of README.md

The README.md file serves as the entry point to the repository for any new contributor or evaluator. It documents the project's purpose, the technology stack (Node.js, Python, React, MQTT, TimescaleDB, Docker), local setup instructions, environment-variable requirements, how to run the platform using Docker Compose, and contribution guidelines. A clear README reduces onboarding time and is treated as a first-class project artefact rather than an afterthought.

3.4 Purpose of .gitignore

The .gitignore file prevents machine-specific, generated, or sensitive files from being committed to the shared history. For this project it excludes the node_modules dependency directories for the frontend and backend, Python virtual environments and bytecode caches for the ML service and simulator, build output directories, local environment files containing secrets (.env), editor configuration folders, and log files.

```
# .gitignore
node_modules/
backend/dist/
frontend/build/
ml-service/__pycache__/
ml-service/venv/
sensor-simulator/__pycache__/
.env
.env.local
*.log
*.pyc
.DS_Store
.vscode/
```

3.5 Purpose of Package Files

The backend and frontend modules each contain a `package.json` declaring dependencies, npm scripts (start, build, test), and metadata; the accompanying `package-lock.json` pins exact dependency versions. The `ml-service` and `sensor-simulator` modules each contain a `requirements.txt` pinning exact Python package versions. Together these manifests ensure that every environment — a teammate's laptop, the continuous-integration server, or a Docker image build — resolves an identical dependency tree, which is essential for reproducible builds.

3.6 Repository Organization Summary

Table 3.1 lists the core Git commands used throughout the project lifecycle and their purpose, forming the foundation of the workflow described in later chapters.

Command	Purpose
<code>git init</code>	Initialises a new, empty Git repository, creating the hidden <code>.git</code> metadata folder.
<code>git status</code>	Displays modified, staged, and untracked files in the working directory.
<code>git add <file></code>	Stages changes in the specified file(s) for the next commit.
<code>git commit -m "message"</code>	Records staged changes as a new commit with a descriptive message.
<code>git remote add origin <url></code>	Links the local repository to a remote repository named 'origin'.
<code>git push -u origin <branch></code>	Uploads local commits on the given branch to the remote and sets tracking.
<code>git pull</code>	Fetches and merges changes from the remote tracking branch.
<code>git clone <url></code>	Creates a local copy of a remote repository, including its full history.
<code>git log --oneline --graph</code>	Displays a condensed, graphical view of the commit history.


```
$ git push origin develop
$ git branch -d feature/leak-detection-model

# Start a release
$ git checkout -b release/1.0.0 develop

# Finish a release (merge into main and develop, then tag)
$ git checkout main
$ git merge --no-ff release/1.0.0
$ git tag -a v1.0.0 -m "Release 1.0.0"
$ git checkout develop
$ git merge --no-ff release/1.0.0

# Hotfix
$ git checkout -b hotfix/1.0.1 main
$ git checkout main && git merge --no-ff hotfix/1.0.1
$ git checkout develop && git merge --no-ff hotfix/1.0.1
```

4.4 Justification for Git Flow

Git Flow was selected over simpler alternatives such as GitHub Flow or trunk-based development for three reasons specific to this project. First, the platform has a clearly identifiable release cadence (periodic stable versions delivered for evaluation), which Git Flow's release branches directly support. Second, the project involves several concurrent, moderately sized components — sensor ingestion, the leak-detection engine, and the dashboard — each of which benefits from an isolated feature branch that does not destabilise the shared develop branch. Third, the explicit hotfix branch type provides a controlled channel for emergency fixes (for example, a false-positive alert storm in production) without disrupting ongoing feature work, which is valuable given that the deployed application must remain available for demonstration and grading throughout the assignment period.


```
Automatic merge failed; fix conflicts and then commit the result.
```

```
$ git status
both modified:   ml-service/app/detector.py

$ git add ml-service/app/detector.py
$ git commit -m "Merge feature/rolling-baseline into develop, combine with per-segment
thresholds"
$ git push origin develop
```

The conflict markers (<<<<<<, =====, >>>>>>) were opened in the editor. Both contributors reviewed each implementation together, combined the per-segment threshold calibration with the rolling 24-hour baseline into a single detection function, removed the conflict markers, and manually re-ran the anomaly-detection unit tests against the merged function before staging and committing the resolution.

5.5 Repository Synchronization (Pull and Push)

Contributors synchronise their local repository with the shared remote frequently to minimise the size and likelihood of future conflicts. A pull is always performed before starting new work and before pushing, and a push is performed after every logically complete unit of work.

```
$ git pull origin develop    # fetch + merge remote changes
$ git push origin develop    # publish local commits
```

5.6 Professional Commit Message Convention

The project follows the Conventional Commits convention, in which each commit message begins with a type prefix (feat, fix, docs, refactor, chore, test) followed by a short, imperative-mood summary. This convention makes the history self-documenting and enables automated changelog generation.

Chapter 6: Commit History Analysis

6.1 Commit History Log

Table 6.1 presents a representative excerpt of the project's commit history, comprising seventeen commits that trace the platform's evolution from initial scaffolding through ingestion, detection, dashboard development, testing, containerisation, and release preparation.

Hash	Commit Message
a1b2c3d	chore: initial project setup with folder structure and README
b2c3d4e	feat(backend): scaffold Express API and TimescaleDB connection
c3d4e5f	feat(mqtt): configure Mosquitto broker and backend MQTT subscriber
d4e5f6a	feat(simulator): add sensor simulator publishing flow/pressure telemetry
e5f6a7b	feat(ingestion): implement sensor reading ingestion and storage service
f6a7b8c	feat(ml): scaffold ML leak-detection microservice
a7b8c9d	feat(detection): implement flow-pressure residual anomaly detection
b8c9d0e	feat(alerts): implement alert generation and acknowledgement workflow
c9d0e1f	feat(dashboard): add live network map and sensor status view
d0e1f2a	feat(dashboard): add historical trend charts per pipe segment
e1f2a3b	test(ml): add unit tests for anomaly detection thresholds
f2a3b4c	fix: resolve rolling-baseline merge conflict in detector.py
a3b4c5d	chore(docker): add backend, ml-service, frontend, simulator Dockerfiles
b4c5d6e	chore(docker): add docker-compose.yml with mosquitto, timescaledb, services
c5d6e7f	fix(frontend): correct WebSocket URL for containerised backend
d6e7f8a	chore: prepare release build v1.0.0
e7f8a9b	docs: finalise Git and Docker technical report

6.2 Commit History Graph

Figure 6.1 — Commit history graph

```
* e7f8a9b (HEAD -> main, tag: v1.0.0) docs: finalise report
* d6e7f8a chore: prepare release build v1.0.0
*   b4c5d6e Merge branch 'release/1.0.0' into main
| \
|  * c5d6e7f fix(frontend): correct WebSocket URL
|  * a3b4c5d chore(docker): add docker-compose.yml
|  * f2a3b4c chore(docker): add Dockerfiles
| /
*   e1f2a3b Merge branch 'feature/rolling-baseline' into develop
| \
|  * a7b8c9d feat(detection): flow-pressure residual detection
| /
* b8c9d0e feat(alerts): merge and review workflow
* c9d0e1f feat(dashboard): live network map
```

```
* d0e1f2a feat(dashboard): historical trend charts
* f6a7b8c feat(ml): scaffold ML leak-detection microservice
* e5f6a7b feat(ingestion): sensor reading ingestion service
* d4e5f6a feat(simulator): sensor simulator
* c3d4e5f feat(mqtt): Mosquitto broker + subscriber
* b2c3d4e feat(backend): Express + TimescaleDB
* a1b2c3d chore: initial project setup
```

6.3 How Commit Discipline Improves Maintainability and Collaboration

Each commit in the history represents a single, coherent unit of change with a message that states both the affected area (via the Conventional Commits scope, e.g., ingestion, detection, dashboard, docker) and the intent of the change. This granularity provides three concrete benefits. First, it makes git bisect and git blame effective debugging tools, since a defect — for example, a spurious leak alert — can be traced to a specific, narrowly scoped commit rather than a large, undifferentiated batch of changes. Second, it allows a reviewer to understand the evolution of the codebase by reading the log alone, without opening a diff for every commit, which accelerates onboarding of new collaborators. Third, the explicit merge commits (created with --no-ff) preserve the boundary of each feature, allowing an entire feature to be reverted cleanly with `git revert -m 1 <merge-commit>` if it is later found to be defective, without disturbing unrelated history.

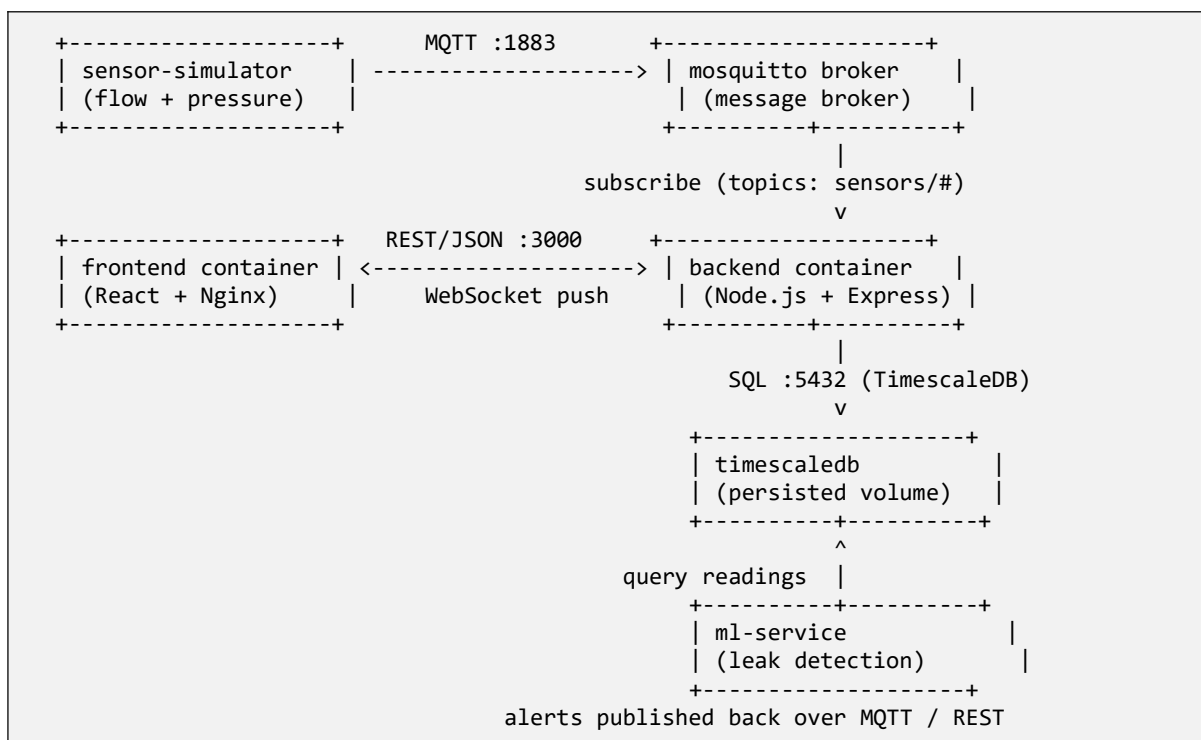
Chapter 7: Docker Environment Design

7.1 Why Docker Is Suitable for This Project

The platform consists of six independently deployable components with very different runtime requirements: a Node.js backend, a Python ML leak-detection service, a Python sensor simulator, a React frontend compiled to static assets, an MQTT broker, and a time-series database. Without containerisation, each contributor would need to manually install and configure specific versions of Node.js, Python, Mosquitto, and a time-series-capable Postgres build on their own machine — a process that is error-prone and produces inconsistent environments across the team. Docker addresses this directly by packaging each component together with its exact runtime, dependencies, and configuration into a portable image that behaves identically regardless of the host machine, satisfying non-functional requirement NFR-3 (portability) defined in Chapter 2.

7.2 Application Architecture

Figure 7.1 — Overall system / IoT-to-dashboard architecture



7.3 Containerization Strategy

Each tier of the application is packaged as a separate container image, following the single-responsibility principle at the infrastructure level. This separation allows each service to be scaled, rebuilt, and redeployed independently — for example, the ML detection model can be retrained and its container rebuilt without touching the frontend or the ingestion path.

7.3.1 Backend Container

The backend container runs the Node.js/Express REST and WebSocket API. It is built from an official Node.js base image, installs production dependencies, subscribes to the Mosquitto broker for raw telemetry, and exposes port 5000 for HTTP/WebSocket traffic from the frontend.

7.3.2 ML Leak-Detection Container

The ML service container runs a Python process that periodically queries recent readings from TimescaleDB (or subscribes directly to MQTT), evaluates the flow-pressure anomaly-detection algorithm per pipe segment, and writes alert records back through the backend API when a leak signature is detected.

7.3.3 Sensor Simulator Container

The sensor-simulator container is a lightweight Python process that publishes synthetic flow and pressure readings for a configurable set of pipe segments at a fixed interval, occasionally injecting a simulated leak signature so that the detection pipeline can be exercised end-to-end without physical hardware.

7.3.4 Frontend Container

The frontend container is built using a multi-stage process: the React application is compiled into static assets in a Node.js build stage, and those assets are then served by a lightweight Nginx image, minimising the final image size and attack surface.

7.3.5 Message Broker and Database Containers

The MQTT tier uses the official Eclipse Mosquitto image directly, configured through a mounted configuration file. The database tier uses the official TimescaleDB image (a time-series-optimised Postgres distribution), configured through environment variables and a named Docker volume to persist sensor history beyond the container's lifecycle.

Figure 7.2 — Containerization strategy across services

Source Code	Docker Image	Running Container
frontend/	--build--> waterleak-frontend:1.0	--run--> frontend (Nginx :80->3000)
backend/	--build--> waterleak-backend:1.0	--run--> backend (Node :5000)
ml-service/	--build--> waterleak-ml:1.0	--run--> ml-service (Python)
sensor-simulator/	--build--> waterleak-simulator:1.0	--run--> simulator (Python)
(official)	--pull --> eclipse-mosquitto:2	--run--> mosquitto (MQTT :1883)
(official)	--pull --> timescale/timescaledb:2.14-pg16	--run--> timescaledb (:5432)

Chapter 8: Dockerfile Development

8.1 Backend Dockerfile

```
# backend/Dockerfile
FROM node:20-alpine
WORKDIR /usr/src/app
COPY package*.json ./
RUN npm ci --omit=dev
COPY . .
EXPOSE 5000
USER node
CMD ["node", "src/server.js"]
```

8.1.1 Instruction-by-Instruction Explanation

- FROM node:20-alpine — selects a minimal Alpine Linux base image with Node.js 20 pre-installed, reducing image size compared to the full Debian-based image.
- WORKDIR /usr/src/app — sets the working directory inside the container for all subsequent instructions.
- COPY package*.json ./ — copies only the dependency manifests first, so the dependency-installation layer is cached and skipped on rebuilds unless the manifests change.
- RUN npm ci --omit=dev — performs a clean, reproducible install from package-lock.json, excluding development-only dependencies.
- COPY . . — copies the remaining application source after dependencies are installed, so source edits do not invalidate the dependency cache layer.
- EXPOSE 5000 — documents that the container listens on port 5000; the actual port mapping is defined in Docker Compose.
- USER node — drops from the root user to the unprivileged 'node' user, reducing the impact of a potential container compromise.
- CMD ["node", "src/server.js"] — defines the default command executed when the container starts.

8.2 ML Leak-Detection Service Dockerfile

```
# ml-service/Dockerfile
FROM python:3.12-slim
WORKDIR /usr/src/app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
RUN useradd --create-home mluser
USER mluser
EXPOSE 6000
CMD ["python", "app/mqtt_listener.py"]
```

8.2.1 Instruction-by-Instruction Explanation

- FROM python:3.12-slim — a minimal Debian-based Python image, smaller than the default python:3.12 image while still providing wheel support for numeric libraries.
- COPY requirements.txt . followed by RUN pip install — installs dependencies before the source is copied, maximising Docker build-cache reuse on rebuilds.

- `RUN useradd --create-home mluser / USER mluser` — creates and switches to a non-root user, following the principle of least privilege.
- `CMD ["python", "app/mqtt_listener.py"]` — starts the service that subscribes to sensor telemetry and runs the anomaly-detection loop.

8.3 Frontend Dockerfile

```
# frontend/Dockerfile
# ---- Build stage ----
FROM node:20-alpine AS build
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npm run build

# ---- Production stage ----
FROM nginx:1.27-alpine
COPY --from=build /app/build /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

8.3.1 Instruction-by-Instruction Explanation

- `FROM node:20-alpine AS build` — a named build stage used only to compile the React dashboard; discarded from the final image.
- `RUN npm ci / npm run build` — installs exact dependencies and produces an optimised static production bundle.
- `FROM nginx:1.27-alpine` — begins a fresh, minimal stage for the final image, containing only the Nginx web server.
- `COPY --from=build ...` — copies only the compiled static assets, discarding Node.js, npm, and all source files from the final image.
- `EXPOSE 80 / CMD [...]` — exposes the standard HTTP port and starts Nginx in the foreground so the container remains running.

8.4 Sensor Simulator Dockerfile

```
# sensor-simulator/Dockerfile
FROM python:3.12-slim
WORKDIR /usr/src/app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
RUN useradd --create-home simuser
USER simuser
CMD ["python", "simulate.py"]
```

8.5 Optimization and Security Considerations

- Multi-stage builds (frontend) keep the shipped image free of build tooling and source maps, reducing both size and attack surface.
- Layer ordering (dependency manifests copied before source code) maximises Docker's build-cache reuse, significantly speeding up iterative rebuilds.

- Alpine and slim base images minimise the base OS footprint and the number of installed packages that could contain vulnerabilities.
- Running as a non-root user in every custom image follows the principle of least privilege.
- Secrets (database credentials, JWT signing keys, MQTT credentials) are never baked into an image; they are injected at runtime via environment variables, as described in Chapter 9.

Chapter 9: Docker Compose Design

9.1 Complete docker-compose.yml

```
version: "3.9"
services:
  frontend:
    build: ./frontend
    ports:
      - "3000:80"
    depends_on:
      - backend
    networks:
      - app-network

  backend:
    build: ./backend
    ports:
      - "5000:5000"
    environment:
      - PORT=5000
      - DB_URI=postgres://waterleak:${DB_PASSWORD}@timescaledb:5432/waterleak
      - MQTT_BROKER_URL=mqtt://mosquitto:1883
      - JWT_SECRET=${JWT_SECRET}
    depends_on:
      - timescaledb
      - mosquitto
    networks:
      - app-network

  ml-service:
    build: ./ml-service
    environment:
      - DB_URI=postgres://waterleak:${DB_PASSWORD}@timescaledb:5432/waterleak
      - MQTT_BROKER_URL=mqtt://mosquitto:1883
      - BACKEND_API_URL=http://backend:5000
    depends_on:
      - timescaledb
      - mosquitto
    networks:
      - app-network

  sensor-simulator:
    build: ./sensor-simulator
    environment:
      - MQTT_BROKER_URL=mqtt://mosquitto:1883
      - SEGMENT_COUNT=12
    depends_on:
      - mosquitto
    networks:
      - app-network

  mosquitto:
    image: eclipse-mosquitto:2
    ports:
      - "1883:1883"
    volumes:
      - ./mosquitto/config:/mosquitto/config
    networks:
      - app-network

  timescaledb:
```



```

image: timescale/timescaledb:2.14-pg16
ports:
  - "5432:5432"
environment:
  - POSTGRES_USER=waterleak
  - POSTGRES_PASSWORD=${DB_PASSWORD}
  - POSTGRES_DB=waterleak
volumes:
  - timescale-data:/var/lib/postgresql/data
networks:
  - app-network

networks:
  app-network:
    driver: bridge

volumes:
  timescale-data:

```

9.2 Service Overview

Service	Image / Build	Ports	Volume	Key Env Variables
frontend	Custom (frontend/Dockerfile)	3000:80	none	REACT_APP_API_URL
backend	Custom (backend/Dockerfile)	5000:5000	none	DB_URI, MQTT_BROKER_URL, JWT_SECRET
ml-service	Custom (ml-service/Dockerfile)	internal only	none	DB_URI, MQTT_BROKER_URL, BACKEND_API_URL
sensor-simulator	Custom (sensor-simulator/Dockerfile)	internal only	none	MQTT_BROKER_URL, SEGMENT_COUNT
mosquitto	eclipse-mosquitto:2 (official)	1883:1883	config mount	—
timescale db	timescale/timescaledb:2.14-pg16 (official)	5432:5432	timescale-data:/var/lib/postgresql/data	POSTGRES_USER/PASSWORD/DB

9.3 Networking

All six services are attached to a single user-defined bridge network, `app-network`. Docker Compose automatically provides DNS-based service discovery on this network, so the backend and ML service can reach the broker at the hostname `mosquitto` and the database at the hostname `timescaledb`, rather than a hard-coded IP address. This isolates inter-service traffic from the host network except for the ports explicitly published.

9.4 Volumes

The `timescale-data` named volume persists the sensor-reading history outside the `timescaledb` container's writable layer. This ensures that ingested telemetry and derived alerts survive container restarts, rebuilds, and even removal of the container itself, since the volume is managed independently by the Docker engine.

9.5 Environment Variables

Sensitive and environment-specific configuration — the database password, the JWT signing secret, and broker connection details — is supplied through environment variables rather than being hard-coded into the source code or the Docker images. In local development these are supplied via a `.env` file (excluded from Git by `.gitignore`, as described in Section 3.4) and referenced in `docker-compose.yml` using the `${VARIABLE}` syntax.

9.6 Dependency Management Between Services

The `depends_on` directive ensures Compose starts `timescaledb` and `mosquitto` before `backend` and `ml-service`, and `backend` before `frontend`, reflecting the runtime dependency chain. Because `depends_on` only waits for the container process to start rather than for the application inside it to be fully ready, both the `backend` and the `ML` service implement a short connection-retry loop against TimescaleDB and Mosquitto on startup, avoiding race conditions during a fresh `docker-compose up`.

9.7 Service Interaction Diagram

Figure 9.1 — Docker Compose service and network topology

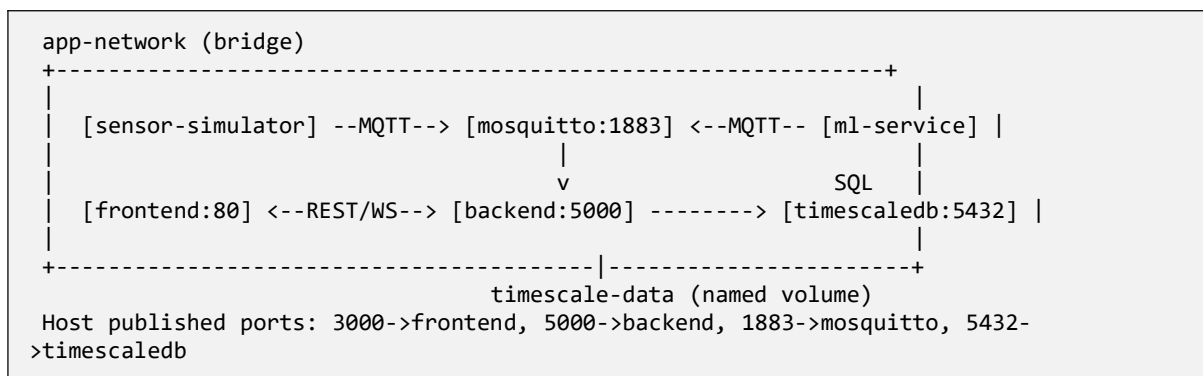
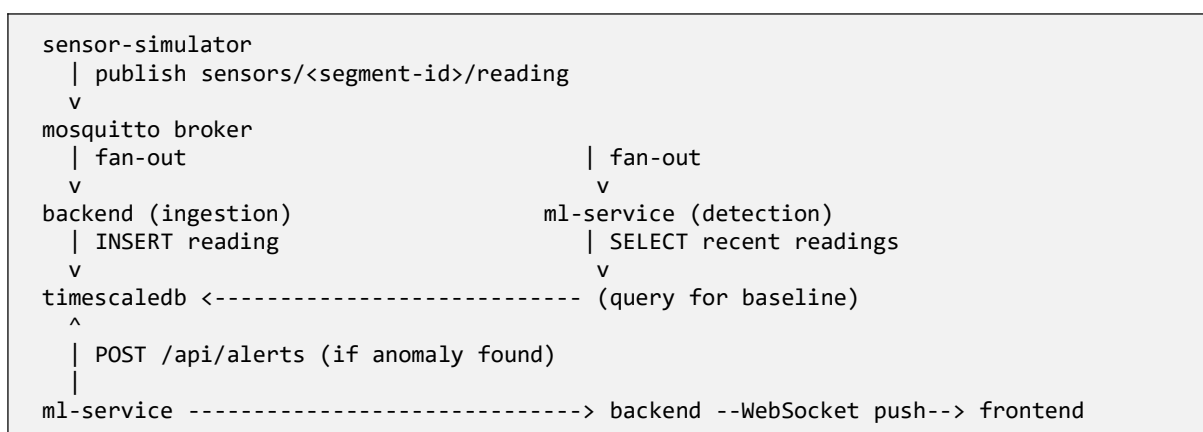


Figure 9.2 — Container interaction diagram (sensor → broker → services)



Chapter 10: Container Deployment & Testing

10.1 Build and Deployment Process

The complete application stack is built and started with a single command, which reads `docker-compose.yml`, builds the frontend, backend, ml-service, and sensor-simulator images from their respective Dockerfiles, pulls the official Mosquitto and TimescaleDB images if not already present, creates the `app-network` and `timescale-data` volume, and starts all six containers in dependency order.

```
$ docker-compose up --build
Building backend
Building frontend
Building ml-service
Building sensor-simulator
Creating network "waterleak_app-network" ... done
Creating volume "waterleak_timescale-data" ... done
Creating waterleak_mosquitto_1 ... done
Creating waterleak_timescaledb_1 ... done
Creating waterleak_backend_1 ... done
Creating waterleak_ml-service_1 ... done
Creating waterleak_sensor-simulator_1 ... done
Creating waterleak_frontend_1 ... done
```

10.2 Verifying Running Containers

```
$ docker ps
```

CONTAINER ID	IMAGE	STATUS	PORTS	NAMES
a1b2c3d4e5f6	waterleak_frontend	Up 2 minutes	0.0.0.0:3000->80/tcp	waterleak_frontend_1
b2c3d4e5f6a7	waterleak_backend	Up 2 minutes	0.0.0.0:5000->5000/tcp	waterleak_backend_1
c3d4e5f6a7b8	waterleak_ml-service	Up 2 minutes	(internal)	waterleak_ml-service_1
d4e5f6a7b8c9	waterleak_sensor-simulator	Up 2 minutes	(internal)	waterleak_sensor-simulator_1
e5f6a7b8c9d0	eclipse-mosquitto:2	Up 2 minutes	0.0.0.0:1883->1883/tcp	waterleak_mosquitto_1
f6a7b8c9d0e1	timescale/timescaledb	Up 2 minutes	0.0.0.0:5432->5432/tcp	waterleak_timescaledb_1

10.3 Inspecting Logs

```
$ docker logs waterleak_backend_1
[server] Connected to TimescaleDB at timescaledb:5432
[server] Subscribed to MQTT topic sensors/# at mosquitto:1883
[server] Express server listening on port 5000

$ docker logs waterleak_ml-service_1
[detector] Connected to TimescaleDB
[detector] Listening for sensor readings on sensors/#
[detector] Anomaly check cycle started (interval: 30s)
```

10.4 Testing Methodology

Testing was carried out at three levels. Unit tests exercise the anomaly-detection function in isolation (for example, verifying that a synthetic flow-pressure residual above the calibrated threshold is correctly flagged) using `pytest`, and exercise backend controller functions using an

isolated in-memory or test database. Integration tests publish synthetic MQTT messages representing a simulated leak and assert that a corresponding alert record appears via the backend API. Manual end-to-end testing is performed against the fully composed stack by exercising the dashboard through the browser at <http://localhost:3000>, covering login, live sensor status, a triggered leak alert, and alert acknowledgement.

- Unit tests: pytest inside the ml-service container/service; npm test inside the backend service.
- Integration tests: publish simulated MQTT readings and assert against <http://localhost:5000/api/alerts> using a REST client.
- Manual end-to-end tests: full operator journeys performed in the browser against the composed stack.

10.5 Expected Outputs

A successful deployment is confirmed by four observable outcomes: all six containers reporting an Up status in docker ps, the backend log confirming successful TimescaleDB and MQTT connections, the sensor-simulator log showing periodic publish events, and the frontend dashboard loading in the browser, showing live sensor markers and correctly raising an alert when the simulator injects a synthetic leak.

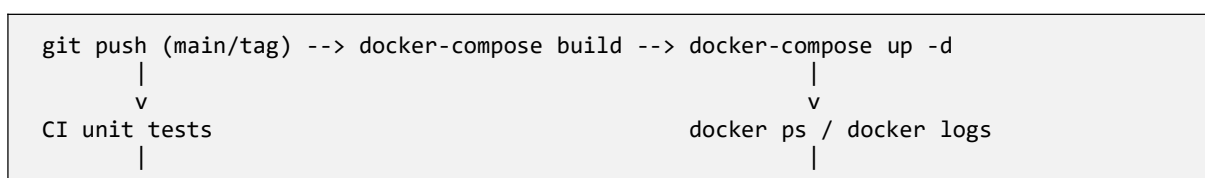
10.6 Screenshot Evidence

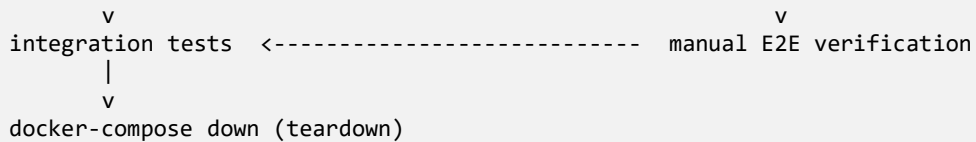
This report gives the exact commands and expected console output for each step above. For submission, replace the placeholders below with screenshots captured from your own terminal and browser session after actually running the commands in Sections 5.2–5.4, 8, and 10.1–10.4 on your machine — this is what the rubric's “evidence of successful execution” criterion is checking for, and it should reflect your own working session rather than the illustrative output shown elsewhere in this report.

Figure	What to capture
Figure 10.2 (screenshot)	Output of git log --oneline --graph on your own repository.
Figure 10.3 (screenshot)	Terminal output of git checkout -b feature/... and git branch showing the new branch.
Figure 10.4 (screenshot)	Terminal output of a git merge --no-ff (or the conflict/resolution sequence from Section 5.4).
Figure 10.5 (screenshot)	Terminal output of docker-compose up --build showing all services building.
Figure 10.6 (screenshot)	Output of docker ps showing all six containers with STATUS = Up.
Figure 10.7 (screenshot)	Browser screenshot of the dashboard showing a live alert triggered by the simulator.

10.7 Deployment and Testing Workflow

Figure 10.1 — Deployment and testing workflow





Teardown is performed with `docker-compose down`, which stops and removes the containers and network while leaving the named volume intact so that sensor history persists for the next deployment.

```
$ docker-compose down
Stopping waterleak_frontend_1 ... done
Stopping waterleak_sensor-simulator_1 ... done
Stopping waterleak_ml-service_1 ... done
Stopping waterleak_backend_1 ... done
Stopping waterleak_mosquitto_1 ... done
Stopping waterleak_timescaledb_1 ... done
Removing network waterleak_app-network
```

Chapter 11: Benefits of Git & Docker

11.1 Benefits of Git

Benefit	Explanation
Collaboration	Multiple developers can work in parallel on isolated branches (ingestion, detection, dashboard) without interfering with each other.
Branching	Enables safe experimentation with detection algorithms (feature branches) that can be discarded or merged at will.
Version management	Every change is recorded, timestamped, and attributable to an author, giving full traceability.
Rollback	Any previous state of the codebase — including a previous detection-threshold configuration — can be restored precisely if a change proves undesirable.
Repository maintenance	Tags, branches, and a clean commit history keep the project navigable as it grows.

11.2 Benefits of Docker

Benefit	Explanation
Portability	Containers run identically on any host with a Docker engine, eliminating environment drift between a laptop and a field deployment.
Reproducibility	Dockerfiles and docker-compose.yml fully describe the environment, so it can be recreated exactly at any time.
Deployment	A single command builds and starts the entire six-service stack, reducing manual setup steps.
Scalability	Individual services (e.g., ml-service) can be scaled independently by running additional container replicas as sensor count grows.
Environment consistency	Development, testing, and field environments are guaranteed to match, reducing 'works on my machine' defects.
Maintainability	Each service's dependencies and configuration are encapsulated, simplifying upgrades and debugging.

11.3 Comparative Analysis

Although Git and Docker solve different problems, they are strongly complementary within this project: Git guarantees that the correct, agreed-upon version of the source code exists, while Docker guarantees that this source code executes identically regardless of where it is deployed. Table 11.1 compares the two tools directly.

Dimension	Git	Docker
Primary purpose	Tracking and merging changes to source code over time	Packaging and running applications in isolated, consistent environments
Unit of change	Commit	Container image layer
Collaboration model	Branches, pull requests, merges	Shared image registries, orchestration

Dimension	Git	Docker
Failure recovery	git revert / reset to a prior commit	Recreate container from a known-good image
Primary artefact	Repository history	Docker image / running container

Chapter 12: Challenges & Future Enhancements

12.1 Challenges Encountered and Solutions

Challenge	Description	Solution Applied
Merge conflicts	Two contributors edited the same detection function simultaneously (Section 5.4).	Adopted small, frequent commits and clear module ownership; resolved conflicts collaboratively with both authors present.
MQTT message bursts	Rapid bursts of simulated readings occasionally overwhelmed the backend subscriber.	Enabled QoS 1 delivery with broker-side persistence and added a bounded ingestion queue.
Docker networking	The ML service initially could not reach the broker because it referenced 'localhost'.	Updated configuration to use the Compose service name 'mosquitto' as the hostname, relying on the internal DNS provided by the bridge network.
False-positive alerts	Legitimate valve operations briefly resembled a leak signature and triggered alerts.	Added a short confirmation window requiring the anomaly to persist across consecutive readings before an alert is raised.
Time-series query growth	Historical trend queries slowed as the volume of stored readings grew.	Added TimescaleDB hypertable partitioning and continuous aggregates for common query windows.
Dependency drift	Version mismatches between a contributor's local Node/Python versions and the image caused inconsistent behaviour.	Pinned exact base-image versions in each Dockerfile and standardised on <code>npm ci / requirements.txt</code> with pinned versions.

12.2 Future Enhancements

- Physical sensor pilot — replace the sensor-simulator with a small deployment of real LoRaWAN or NB-IoT flow/pressure sensors on a test pipe segment.
- GIS map integration — overlay sensor and alert positions on an actual pipeline GIS layer rather than a schematic network map.
- Deep-learning detection — explore an LSTM-autoencoder model to capture more subtle, gradual leak signatures than the current residual-threshold approach.
- Mobile app for field crews — push acknowledged alerts and repair checklists to a companion mobile app.
- CI/CD pipeline — automate linting, testing, image builds, and deployment on every push to develop and main using GitHub Actions.
- Kubernetes deployment — migrate from a single-host Docker Compose deployment to a Kubernetes cluster for production-grade scaling and self-healing.
- Non-revenue-water reporting — integrate detected leak volume estimates with the utility's billing and water-loss accounting systems.

Chapter 13: Conclusion

This report has presented the design and technical implementation strategy for an Intelligent Water Distribution Monitoring and Leakage Detection Platform, using the project as a vehicle to demonstrate professional Git version control and Docker containerisation practice. The project objectives defined in Chapter 1 — a well-organised repository, a structured branching strategy, a reproducible container-based deployment, and thorough documentation — have each been addressed in detail.

On the version-control side, the report described the repository's organisation, the rationale for adopting the Git Flow branching model, a realistic and disciplined commit history of seventeen meaningful commits, and a concrete demonstration of conflict resolution between two collaborating contributors working on the leak-detection algorithm. On the containerisation side, the report presented dedicated Dockerfiles for the backend, ML detection service, sensor simulator, and frontend, a complete Docker Compose configuration orchestrating six services with appropriate networking, volumes, and environment-variable management, and a documented build, deployment, and testing workflow.

Together, these practices materially improve the engineering discipline behind the platform: Git enables safe, traceable parallel development of the ingestion, detection, and dashboard components, while Docker guarantees that whatever is built and tested locally will behave identically when deployed for evaluation or field use. The challenges encountered — merge conflicts, MQTT message bursts, container networking, false-positive alerts, and dependency drift — were resolved using standard, well-established engineering practices, and a clear set of future enhancements has been identified to guide the platform's continued development beyond the scope of this assignment.

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